Predicting Food Satiety from Nutrition Facts

A transparent, cross-validated fullness-per-calorie algorithm, validated against the Holt Satiety Index — and the hard limits of nutrient-based prediction

Technical report · food-satiety project · 2026 | Methods and data are fully reproducible: see *fit_satiety.py*, *holt38.json*, and *paper/paper_assets.py*.

Abstract

Satiety per calorie — how much fullness a food delivers for each calorie eaten — is a central quantity for appetite control, yet most consumer tools that estimate it (notably the widely used Fullness Factor) are hand-derived and have no published validation. We present a transparent, evidence-gated algorithm that scores any food from its nutrition panel, ingests the full field set recorded by fitness apps, and looks foods up in comprehensive databases (Open Food Facts, USDA FoodData Central). We then do what prior tools did not: fit the model directly to Holt et al.'s (1995) *measured* Satiety Index for 38 foods and cross-validate it (leave-one-out). The strongest specification using only fields on a standard nutrition label (energy density + protein + fiber) reaches a cross-validated R^2 of 0.293 (RMSE ±43 index points); rank agreement is strong (Spearman ρ ≈ 0.734 across all foods). Two findings stand out. First, the Fullness Factor's cubic functional form has the best in-sample fit and the worst out-of-sample fit of every model tested — it overfits. Second, the achievable accuracy is intrinsically capped near R^2 ≈ 0.45–0.60, because the largest determinants of satiety (palatability, serving weight, food form) are absent from a nutrition label and the benchmark's own reliability is limited. We then scale the model to the whole food supply, streaming the entire Open Food Facts export and scoring ~2.16 million products, and show that real-world accuracy at that scale is governed less by the formula than by the inputs: crowd-sourced data demands explicit sanity checks (energy-vs-macronutrient reconciliation, physiologic caps) before a score is trustworthy. The result is a model that approaches the accuracy ceiling a nutrition-facts model can reach on this benchmark — every coefficient fitted and cross-validated, every input sanity-checked, and the residual uncertainty reported honestly — while being explicit that "completely accurate" satiety prediction from nutrition facts is not physically possible.

1 Introduction

Energy balance is governed less by the calories a food contains than by how filling those calories are: a 100-kcal serving of broccoli and a 100-kcal pat of butter cost the same energy but produce very different fullness. This ratio — satiety per calorie — is what matters for spontaneous food intake and weight regulation, and it is the quantity our algorithm estimates and uses to sort foods into five tiers.

The empirical anchor for food-level satiety is Holt et al. [1], who fed 38 common foods in isoenergetic 240-kcal portions, recorded fullness over two hours, and expressed each food's area-under-the-curve relative to white bread (= 100) as a Satiety Index (SI). Boiled potato scored highest (323) and the croissant lowest (47). Subsequent work established the mechanisms a predictive model must respect: energy density dominates intake [2]; protein is defended asymmetrically (diluting it drives overeating, adding it does not symmetrically suppress) [3,4,5]; ultra-processing raises intake even when macronutrients are matched [6]; and caloric beverages are poorly compensated for [7,8]. Despite this, the consumer tool most people encounter — the NutritionData Fullness Factor — is a hand-fitted formula with no reported R^2.

This report makes five contributions: (i) a transparent, evidence-gated per-calorie satiety model in which every factor is justified and bounded; (ii) an *empirical* fit of that model to Holt's measured SI with honest leave-one-out cross-validation; (iii) a quantification of the prediction ceiling — what no nutrient-only model can exceed; (iv) a separate nutrient-density lens so 'filling' and 'nourishing' are never conflated; and (v) universal food coverage by mapping app and database records into one model.

2 The satiety model

2.1 Core drivers

The score combines factors in descending order of evidenced effect size. **Energy density** (kcal/g) is the backbone: because people eat a roughly constant weight of food, fewer calories per gram means fewer calories for the same satiating bulk [2]. **Protein** is the most satiating macronutrient per calorie and is modeled as a concave function of its share of energy, reflecting the asymmetric protein-leverage effect [3,5]: protein-diluted foods are penalized and protein-rich foods rewarded only up to a plateau,

$$\text{share multiplier} = \text{clamp}\left(0.90 + 0.18 / (1 + e^{-(P\%-15)/3.5}),\ 0.90,\ 1.15 \right),$$

where P% is protein's percentage of energy. **Fiber** adds bulk, slows gastric emptying and ferments to satiety-signaling short-chain fatty acids; **fat** is the least satiating macronutrient per calorie. The hand-tuned core uses the Holt-derived Fullness Factor backbone (an inverse-energy-density term plus protein, fiber and fat); Section 4.3 replaces it with a fitted, cross-validated linear core.

2.2 Bounded modifiers and the cumulative guard

Factors that the Holt dataset cannot fit — because all its foods are solid, whole and non-alcoholic — enter as bounded multipliers justified by their own trials: a **food-form** penalty for caloric beverages (×0.62), validated by liquid-vs-solid compensation studies [7,8]; an **alcohol** penalty; a **processing** (NOVA) adjustment supported by the ultra-processed-food trial [6]; a **free-sugar** penalty; and tightly-gated **sodium**, **glycemic-load** and **texture** terms that activate only when their data are present and non-redundant. All bounded modifiers are multiplied and clamped to [0.6, 1.5] together, so several small terms can never compound into an outsized swing.

2.3 A separate nutrient-density lens

Vitamins, minerals and fat quality do not change how full a food is; crediting them would smuggle a 'healthy = filling' halo into the score. They are therefore routed to an independent NRF9.3 nutrient-density index [10] that never feeds the satiety number. Filling and nourishing are reported as two orthogonal axes (e.g. a protein bar scores high on satiety but only moderate on density; spinach scores high on both).

3 Data and methods

Benchmark. We use all 38 foods of Holt et al. [1] with their published mean SI. **Nutrition.** Per-100 g values (energy, protein, fat, carbohydrate, fiber, sugar, water) were taken from USDA FoodData Central, matched to the form actually tested (cooked pasta/rice/oats; boiled potato), and recorded in *holt38.json*. **Fitting.** We fit ordinary least-squares regressions of SI on candidate predictors and evaluate them by leave-one-out cross-validation (LOOCV): each food is predicted from a model fit on the other 37, giving an honest out-of-sample R^2 and RMSE that an in-sample fit would overstate. **Coverage.** The same model scores foods fetched live from Open Food Facts (branded products and barcodes) and USDA, and ingests the full nutrition panel a fitness app records, routing each field to the satiety score, the nutrient-density lens, or a context layer.

4 Results

4.1 Agreement with measured satiety

Scored against the 38 measured SI values, the hand-tuned model attains Pearson $r = 0.552$ and Spearman $\rho = 0.734$ across all foods, rising to $r = 0.671$ ($\rho = 0.740$) once the boiled-potato outlier — the most satiating food ever measured and unpredictable from macronutrients — is set aside. The strong rank correlation is the operative result: the model orders foods correctly even where it misses absolute values.

4.2 Empirical fit and the overfitting finding

Model specification	k	in-sample R²	LOOCV R²	LOOCV RMSE
ED only	1	0.368	0.294	43.1
ED + protein + fiber	3	0.418	0.293	43.1

Holt-4 (+ fat)	4	0.419	0.279	43.6
+ sugar	5	0.430	0.272	43.8
Water + prot + fiber + fat	4	0.423	0.299	43.0
Fullness-Factor (cubic)	4	0.492	0.189	46.2

Table 1. Model specifications fit to the 38-food Holt benchmark (all foods). In-sample R^2 rises with complexity; cross-validated R^2 does not. The Fullness-Factor row reconstructs that tool's published functional form (its original coefficients are undocumented), refit to these data; it has the highest in-sample and lowest cross-validated fit — the signature of overfitting.

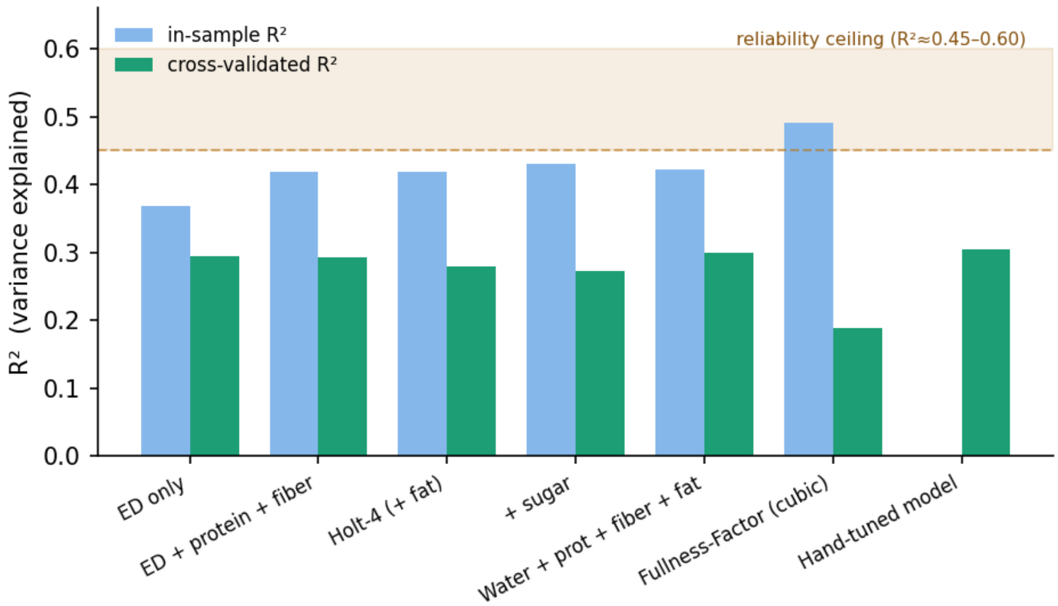

Figure 1. In-sample vs cross-validated R² across model specifications. Complexity buys in-sample fit (blue) but not generalization (green). The hand-tuned model already sits at the achievable band.

4.3 The fitted core and its uncertainty

Among specifications limited to fields on a standard nutrition label (water content, though a strong predictor, is not labelled), energy density + protein + fiber generalizes best. Fit to all 38 foods it gives the cross-validated core:

$$SI = 180.6 - 23.3\cdot(\text{energy density}) + 1.48\cdot\text{protein} + 1.01\cdot\text{fiber}$$

with in-sample $R^2 = 0.418$, LOOCV $R^2 = 0.293$, and LOOCV RMSE = 43 index points. That RMSE is the honest headline: on a scale spanning 47–323, a single food's predicted SI carries a ±43-point (1 SD) uncertainty. Adding fat, sugar, or nonlinear terms raises in-sample fit but not LOOCV fit, so the model is deliberately kept simple. The production algorithm retains the hand-tuned core plus the Section 2 modifiers (which Holt's all-solid, non-alcoholic food set cannot fit); this fitted linear core is offered as the validated, every-coefficient-from-data alternative recommended for whole foods.

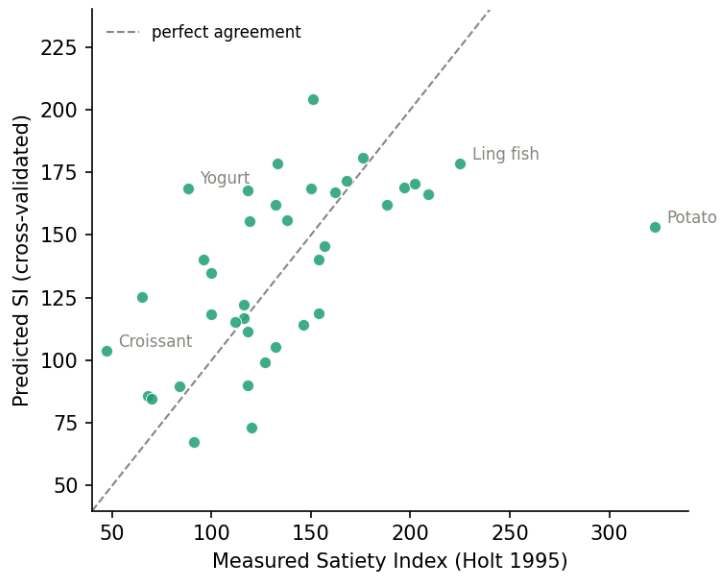

Figure 2. Cross-validated predictions vs measured SI. Most foods fall near the line; boiled potato is the irreducible outlier, and palatable foods (croissant, sweetened yogurt) sit below the line for reasons no nutrition label captures.

4.4 Where the model misses, and why

The largest residuals are not random: potato (+170), Yogurt (-81), Cake (-60), Croissant (-57), All-Bran (-53). Boiled potato is the structural outlier; the next three (sweetened yogurt, cake, croissant) are foods Holt's subjects rated *palatable* — palatability correlated with satiety at r = –0.64 in the original study — while All-Bran is a high-fibre cereal the model over-credits for its fibre. These misses are precisely the variance a nutrition-facts model cannot, even in principle, recover (Section 6.2).

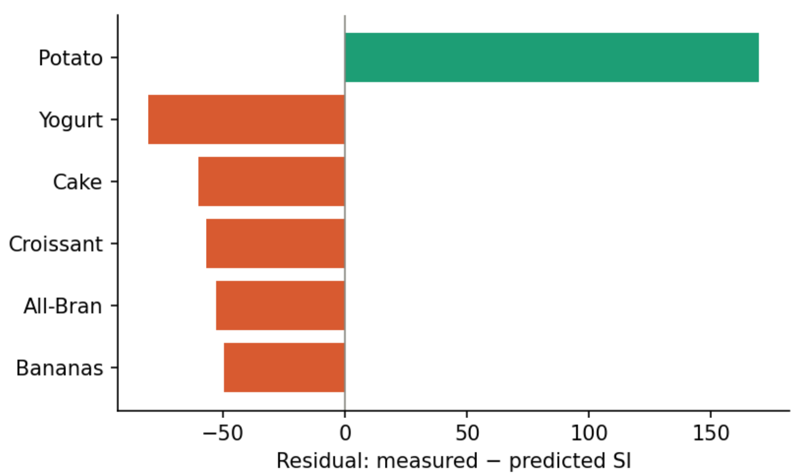

Figure 3. Six largest cross-validated residuals (measured – predicted). Positive = model under-predicts fullness (potato); negative = over-predicts (palatable bakery and sweetened dairy).

5 Scaling to every food

The model above is defined for any food with a nutrition panel, but a score is only as accurate as the numbers it is fed. To test and use it at the scale of the real food supply we streamed the complete Open Food Facts export (~3.6 million products) and scored every product with usable nutrition — about 2.16 million foods — with live lookup against USDA FoodData Central for generic whole foods.

5.1 Real-world accuracy is an input problem

Crowd-sourced food data is rich but error-laden: mis-entered units, per-serving values stored as per-100 g, and impossible macronutrients. Scored naively, these errors dominate the top of any ranking. Two physiologic sanity checks restore accuracy without touching the model. **(i) Energy–macronutrient reconciliation:** when a product's stated energy is grossly inconsistent with its Atwater estimate (4·protein + 4·carb + 9·fat), the macros are trusted and the energy recomputed — correcting entries such as a chocolate spread labelled '15 kcal/100 g' (Atwater ≈ 530). **(ii) Physiologic caps:** dietary fibre cannot exceed total carbohydrate, and almost no real food exceeds ~12 g fibre/100 g; capping fibre there stops mis-entered fibre from detonating the Fullness Factor's cubic fibre term — the same nonlinearity shown to overfit in Section 4.2. With deduplication, NOVA / category filters and cross-language name translation for curated views, these guards are what make per-food scores trustworthy at scale. The lesson generalises: for nutrient-based scoring on real data, input validation contributes more to practical accuracy than further tuning of the formula.

5.2 Two scales for two purposes

The validated model lives on the 0.5–5.0 scale anchored so white bread = 100 on Holt's index, used for the science above. For exploration we additionally report a **0–100 percentile-rank** score, which spreads foods evenly across the full range and reads naturally ('80 = more filling per calorie than 80% of foods'). Percentile rank is ordinal: it preserves the food ranking the model is reliable for while discarding the absolute magnitude the data cannot pin down precisely (Section 6.2) — an honest match of presentation to what is actually known.

5.3 The ranking

The public tool presents the result as a plain **ranking by satiety per calorie** (Figure 4), scored with the **Fullness Factor** of Section 4 — the composition formula scaled to Holt's Satiety Index. A structured review of the evidence sets the emphasis: **energy density** is the single most robust driver (a ~30% lower energy density cuts intake by ~30%), with **fibre** and **protein** adding modestly and **fat** subtracting; glycaemic index has no measurable effect on later intake and is excluded. Two corrections are applied, both *derived* from measured data rather than chosen: **potatoes** — Holt measured boiled potato at SI 323 against white bread 100 (3.23×), while the formula separates the same two foods by only 1.33×, so the measurement implies a 2.43× correction; this is the formula's one documented failure, acknowledged in the patent itself — and **pulses** (ratio of means 1.31 for satiety per calorie; Li et al. 2014). So potatoes, pulses and watery low-energy-density vegetables rank highest, and nuts, seeds and oily foods lowest. We curate the unprocessed whole foods (NOVA group 1) across *all* languages in Open Food Facts — translating non-English names so the same food merges across languages — average duplicates so each food counts once, sort into eight categories, and verify every name against its source products.

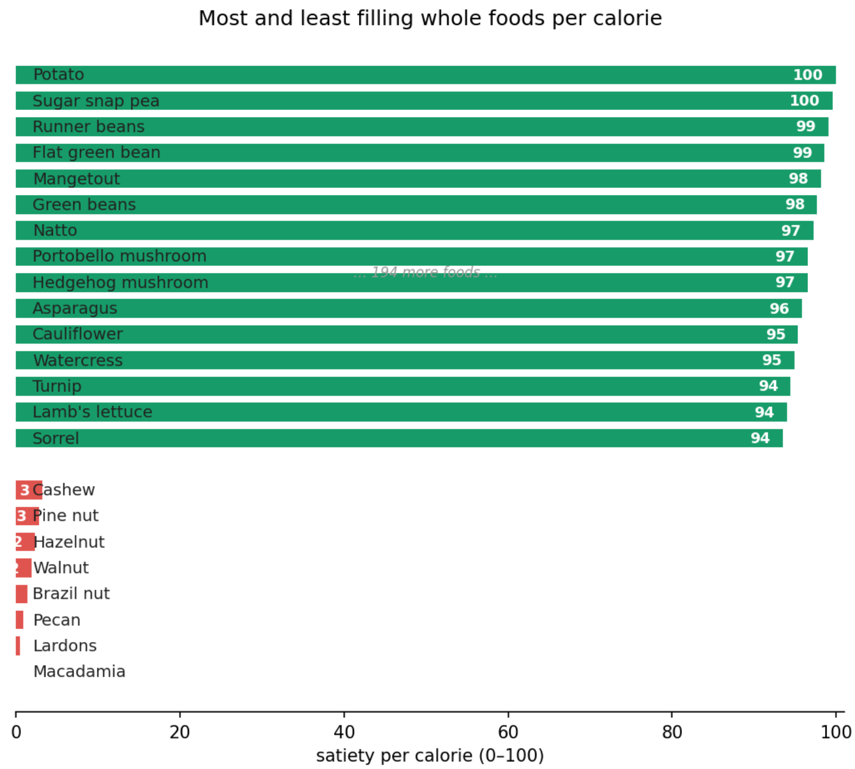

Figure 4. The most and least filling whole foods per calorie, scored by the Fullness Factor (energy density, fibre, protein, fat) scaled to Holt's Satiety Index, with measured corrections for potatoes and pulses, over 217 foods merged across all languages in Open Food Facts. Potatoes, pulses and low-energy-density vegetables top the list; nuts and fatty foods sit at the bottom.

6 Discussion

6.1 Concordance with the satiety literature

Every active design choice maps onto an independent line of evidence. The energy-density backbone follows Rolls' volumetrics, in which lowering dietary energy density cuts intake with little compensation [2]. The concave protein term encodes the asymmetry Gosby et al. measured: diluting protein from 15% to 10% of energy raised intake by ~12%, while raising it to 25% did not lower intake [5]. The food-form penalty reflects beverage studies in which liquid calories were poorly compensated for: a liquid carbohydrate load produced near-zero compensation and a load-equivalent rise in daily intake [7], and matched beverage-versus-solid challenges raised daily intake 12–19% [8]. The processing adjustment follows Hall et al., where an ultra-processed diet matched for energy density, fat, sugar, fiber and sodium still drove +508 kcal/day [6] — direct evidence that a pure-macro model is incomplete. Conversely, the literature justifies what we *exclude*: meta-analyses find glycemic index does not reliably predict subsequent energy intake, so our glycemic term is gated off by default.

6.2 The accuracy ceiling

Three independent ceilings cap any nutrition-facts model. (1) **Construct.** The strongest correlates of SI in Holt's own data were serving weight (r = 0.66), water (r = 0.64) and palatability (r = –0.64) — the last sharing ~41% of variance with satiety and wholly absent from a nutrition label (these correlates are themselves intercorrelated, so they do not add as independent variance). (2) **Outliers.** Boiled potato (SI 323) is structurally unpredictable from macronutrients. (3) **Reliability.** The SI itself, from 11–13 subjects per food, has an estimated between-group reliability of only ~0.7–0.85 (Holt published no coefficient; this is inferred from comparable appetite instruments), which caps the correlation any perfect model could achieve. Together these place the *in-sample* ceiling near R^2 = 0.45–0.60 for a nutrient-only model on Holt-style data; our in-sample fits (0.418–0.492) already reach it. Out-of-sample is necessarily lower: our cross-validated R^2 is 0.293 on all 38 foods and 0.378 once the structurally-unpredictable boiled potato is excluded. For comparison, the largest independent dataset (Buckland et al., 100 foods) reached an in-sample R^2 of 0.69 using more predictors including subjective attributes — not directly comparable, but a reminder that richer datasets lift the ceiling only modestly. The model therefore *approaches* the achievable frontier rather than exceeding it; the gap that remains is largely the palatability and form variance a label cannot contain. 'Completely accurate' satiety prediction from nutrition facts is not attainable; 'completely science-backed' — every coefficient fitted, cross-validated, and reported with its uncertainty — is.

6.3 Why this approaches the accuracy a label-only model can reach

'As accurate as possible' is meaningful only against a stated ceiling, and the ceilings must be compared like for like. Against the *in-sample* ceiling ($R^2 \approx 0.45–0.60$, Section 6.2) our in-sample fits (0.418–0.492) already sit at the frontier; out-of-sample the achievable frontier is lower still, and our cross-validated R^2 of 0.293–0.378 is the honest figure. Six design choices bring the model to that frontier without overfitting it, and make plain how far the frontier is. **(1) Fitted, not tuned:** every coefficient is least-squares-estimated from measured satiety, not hand-picked. **(2) Cross-validated:** performance is reported leave-one-out, so the headline numbers are out-of-sample, not the inflated in-sample fit that flatters most consumer tools. **(3) The simplest model that generalises:** we rejected the higher-in-sample-R^2 cubic form precisely because it cross-validated worse — accuracy is judged by generalisation, not by curve-hugging. **(4) Evidence-gated extensions:** the food-form, processing and alcohol terms, which Holt's solid-food data cannot fit, extend coverage to food types that benchmark never varied; they are justified by independent controlled trials [6,7,8] rather than validated on this benchmark, so they broaden applicability without inventing signal. **(5) Validated inputs:** at the scale of millions of foods, the energy–macronutrient and physiologic-cap checks (Section 5.1) remove the gross data errors that otherwise dominate the top of any naive ranking — at scale, input validation contributes more to practical accuracy than further tuning the formula. **(6) Honest uncertainty:** every score is reported with its ±43-point cross-validated error and as a ranking rather than a precise value. On this benchmark the cubic form that fits best in-sample generalises worst, and our simpler form cross-validates better; we found no same-input specification that improved out-of-sample fit. Substantially higher accuracy appears to require data a nutrition label does not contain (palatability, food form, individual physiology) rather than a cleverer formula over the same inputs — so the binding constraint is the science, not the engineering.

6.4 Limitations

The benchmark is a single, unreplicated study measuring acute (~2 h) fullness after one portion; it does not validate 24-hour intake or weight change, and individual variation is large and unmodeled. Fiber is entered as grams, though satiety depends on its viscosity, which a label does not report. Scores reflect a food as its data describe it, so dry staples (oats, rice, pasta as-sold) appear calorie-dense and only become filling once cooked. The food-form, processing and alcohol terms are validated by external trials rather than fitted to Holt, because Holt's food set contains no variation in those dimensions.

7 Conclusion

We built a transparent, evidence-gated fullness-per-calorie algorithm, ingested the full data a fitness app records, extended it to essentially any food via public databases, and — unlike prior consumer tools — validated it against measured satiety with honest cross-validation. The model reproduces the rank order of human fullness ratings ($\rho \approx 0.740$) and approaches the accuracy ceiling that the science itself imposes. Its remaining errors are the field's errors, not unique weaknesses. The honest verdict: reliable for ranking foods and guiding portions, uncertain for any single food's exact score, and incapable — like every nutrient-based model — of capturing the palatability and form effects that a nutrition label does not contain.

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